2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.306 Internet Applications Engineering

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

This question is about analysing web site performance.

1.a Web Development tools and Waterfall diagrams are particularly important for analysing and improving the performance of web sites.

i. The following is an abstract representation of the vertical bar that is used to show the 7 timings that make up retrieving each web resource, as measured by Google Developer Tools. Replicate this diagram in your answer booklet. Label each box (A-G) with the correct technical term. For each term, provide a short reasoned explanation of why each source of delay can occur. *You can assume this is a non-encrypted connection.*

[9 marks]

ii. The following web resources were timed as they were downloaded by Google Developer Tools. The performance of each request is broken down into the 7 timings from the previous question in the table below (all times are in milliseconds). *You can assume rows with missing numbers are utilising existing TCP connections. The service is transatlantic (over 8 hops away), the largest content item is 2.0MB in size.*

	A (ms)	**B** (ms)	**C** (ms)	**D** (ms)	**E** (ms)	**F** (ms)	**G** (ms)
Request 1	5.12	5.91			2.01	233.21	1250
Request 2	3.95	7.01			0.88	38.01	4.36
Request 3	4.13	7.30			0.087	606.90	1.81
Request 4	0.72	2.09	141.35	64.57	0.095	233.38	0.96

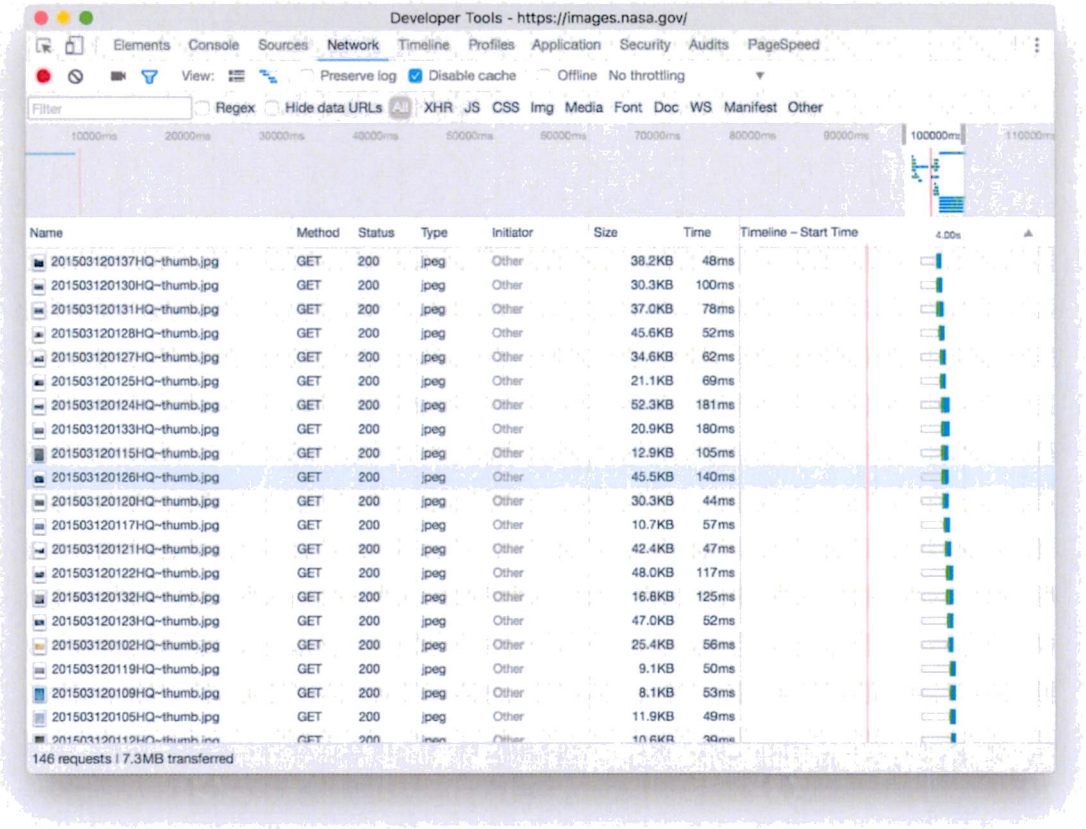
For each of the 4 resources (rows) provide your interpretation of the kind of resource being downloaded. Based on the performance figures available briefly state how good you believe the performance to be. You should include your rationale for this using data from the table above in your answer.

[9 marks]

Question 1 continues on next page...

Question 1 continued…

iii. Look carefully at the waterfall diagram below. You will notice a trend in behaviour affecting the 'white bar' and load time of each resource. Clearly explain the trend and the features you can observe. Provide a detailed explanation as to the cause. Suggest 3 ways in which you could help mitigate this problem.



[7 marks]

Total 25 marks

Question 2

This question is about responsive web design.

2.a When using responsive web design to create websites for a range of devices, why is a fixed canvas size unsuitable?

[1 mark]

2.b Explain the *mobile first* approach and how it is used in responsive web design.

[2 marks]

2.c Fluid grids and media queries represent two of the key principles of responsive web design. Describe the third principle, and explain how it supports a responsive web design approach.

[2 marks]

2.d Your employer has asked you to investigate how news content produced by the company can easily be consumed by desktop and mobile users. The company currently has a website that has been designed for desktops and is not mobile friendly. You must therefore consider whether to recommend that the company produces a responsive version of their website or creates a native mobile application. Discuss two advantages and two disadvantages of each approach.

[4 marks]

2.e The following code snippet is a single media query for a mobile device that implements a 12 column fluid grid.

```
@media only screen and (max-width: 425px) {
    .col-m-0 {width: 0; display: none;}
    .col-m-1 {width: 10%;}
    .col-m-2 {width: 15%;}
    .col-m-3 {width: 20%;}
    .col-m-4 {width: 25%;}
    .col-m-5 {width: 30%;}
    .col-m-6 {width: 40%;}
    .col-m-7 {width: 50%;}
    .col-m-8 {width: 60%;}
    .col-m-9 {width: 70%;}
    .col-m-10 {width: 80%;}
    .col-m-11 {width: 90%;}
    .col-m-12 {width: 100%;}
}
```

Question 2 continues on next page…

Question 2 continued...

i. Explain what is wrong with the code snippet and how it could be resolved.

[4 marks]

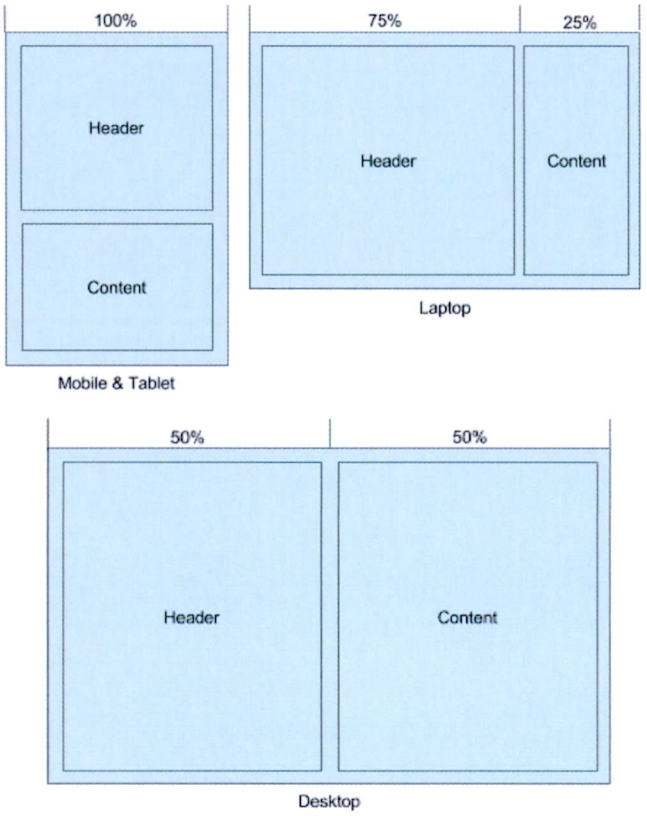
ii. How would you improve this method to cater for other device sizes, such as tablets, laptops and desktops?

[2 marks]

iii. In the previous code snippet 12 columns have been defined for the fluid grid. Is this number of columns necessary for a responsive website to work correctly? Provide an explanation with your answer.

[2 marks]

iv. You are now required to put the fluid grid code (from parts i & ii) into use, and consider two div elements `header` and `content` which are to be arranged differently on device sizes according to the following sketches. Provide the class names for each div element.



[4 marks]

Question 2 continues on next page...

Question 2 continued…

v. Using the sketches on the previous page as examples, sketch the layout that the following div elements would create. You should Include percentages in your sketches (but note that the sketches themselves are **not** required to be drawn to scale).

```
<div id="header" class="col-m-12 col-t-12 col-l-12 col-d-12"></div>

<div id="menu" class="col-m-0 col-t-12 col-l-12 col-d-12"></div>

<div id="content" class="col-m-12 col-t-12 col-l-12 col-d-12">

      <div id="content-main" class="col-m-12 col-t-12 col-l-9 col-d-
      6"></div>

      <div id="content-additional" class="col-m-12 col-t-12 col-l-3
      col-d-3"></div>

</div>
```

[4 marks]

Total 25 marks

Question 3

This question is about the accessibility of mobile and web applications.

3.a In the context of mobile application and web development, how would you define 'accessibility'? Who does it benefit?

[1 mark]

3.b List the four forms of impairment that should be considered when creating an accessible website. For each, describe how they might impact a person's use of the website.

[6 marks]

3.c The following HTML code is an extract from a website that incorporates a range of accessibility features.

```
<head>
<html lang="en">
<title>Lancaster University - Home Page</title>
</head>
<body>
...
<img src="res/lancaster-logo.jpg" alt="Lancaster University Logo"
tabindex="1">
...
<form>
<label for="name">Name:<br>
<input id="name" placeholder="Joe Bloggs" tabindex="2"><br>
<label for="age">Age:<br>
<input id="age" type="number" placeholder="Age" tabindex="3"><br>
<label for="date">Date:<br>
<input id="date" type="date" tabindex="4"><br>
<input id="submit" type="submit" tabindex="5">
</form>
...
</body>
```

i. Study the extract and identify the various ways in which the website has been implemented to maximise accessibility.

[6 marks]

ii. For each accessibility feature that you identify, explain the form(s) of impairment (listed in your answer to part b) that it is designed to address.

[6 marks]

Question 3 continues on next page…

Question 3 continued…

3.d The website owner is considering introducing additional accessibility features that make use of the hardware built into modern smartphones and/or smartwatches. Suggest three additional accessibility features that could utilise the hardware of smartphones and smartwatches and explain, by means of an example, how each would increase the accessibility of a website or application.

[6 marks]

Total 25 marks

Question 4

The following question is about HTTP/2 and web architectures.

4.a Describe three deficiencies with HTTP/1.x and explain how each can be addressed in terms of current web performance best practice.

[7 marks]

4.b Explain how HTTP/2 has been designed to improve web performance. Describe four techniques that HTTP/2 uses and explain how each of them achieves an improvement in performance.

[8 marks]

4.c The BBC have historically made significant use of caching to ensure their infrastructure is capable of handling millions of website views per day.

i. Explain the challenges with using infrastructures that make significant use of caching when building interactive web experiences.

[2 marks]

ii. You have been asked to design a system architecture and propose a set of suitable technologies for building an interactive sports website that includes football goal events and news updates during a live game. Outline how you might achieve this based on your understanding of web architectures. Your answer should include an overview of the system architecture and the technologies you believe would play a key role, including a rationale for their selection. You should also consider how you would deploy these technologies and the computational resources required to provide the service.

[8 marks]

Total 25 marks

---End of Paper---